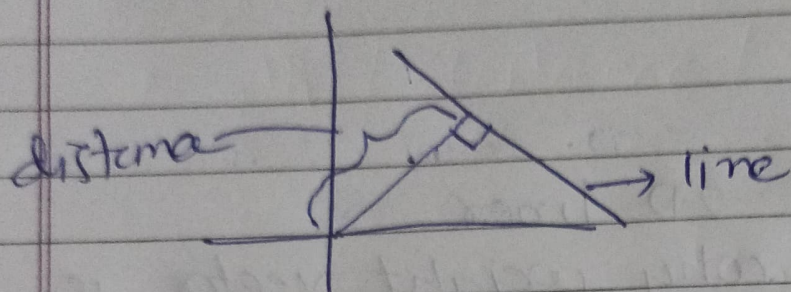


Maths for ML

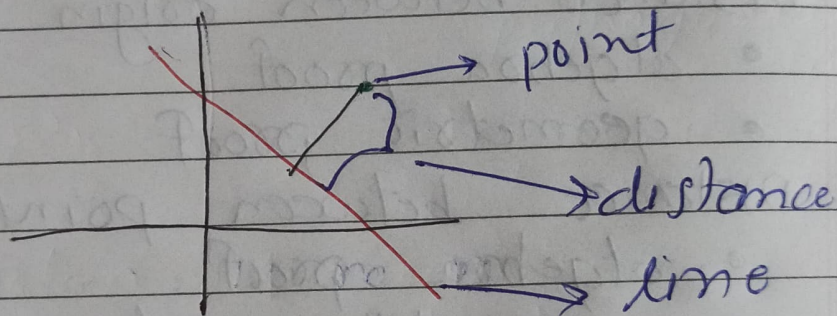
L3 :- Halfspace & distance

- * Recap
- * Shifting 2D lines
- * proof of why weight vector is always perpendicular to line or hyperplane
- * Distance between origin & a line
 - algebra proof
 - geometric proof
- * Distance between point & line
 - algebra proof
 - geometric proof
- * Halfspaces.

distance betⁿ origin & line



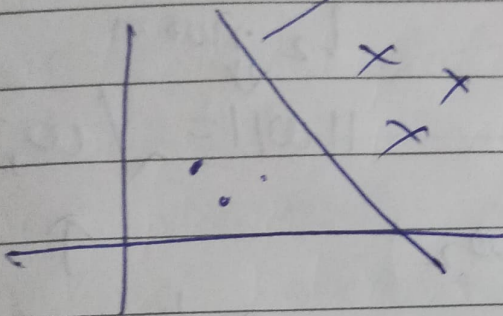
distance betⁿ point & line



see cap

ML needs dataset
dataset contains features & target
gives

ML goal: - finding line



$$y = mx + c$$

$$w_1 x_0 + w_2 \overset{y}{x} + w_3 = 0$$

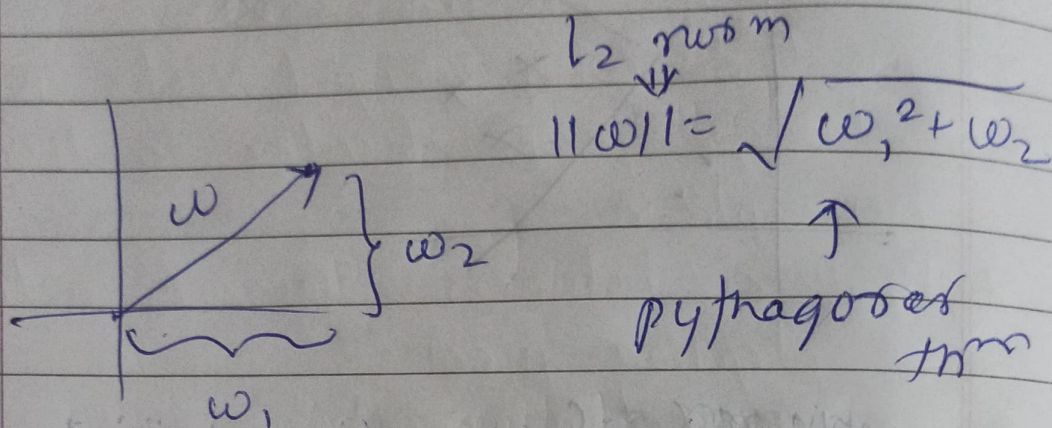
$$y = -\frac{w_1}{w_2} x - \frac{w_3}{w_2}$$

$\downarrow$
m

$\downarrow$
c

vector $\rightarrow [1, 7, 3]$

length / distance / magnitude of
norm $\rightarrow$ all are same

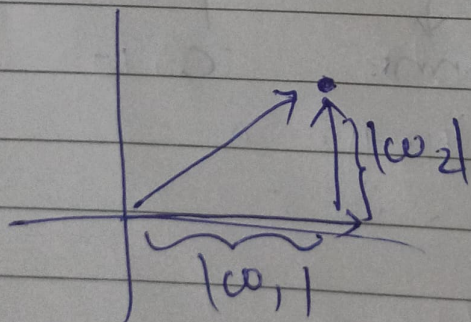


Manhattan ℓ_1 norm

$$\|w\| = |w_1| + |w_2|$$

$\Uparrow$

absolute value of w_1



$$w = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \bar{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$

dot product

$$w^T \bar{x} = 2x + 3y$$

where is dot product is used?

Dot product gives similarity
betw 2 vectors.

high value of dot product
means high similarity

$$A = \begin{bmatrix} 30 & 20 & 11 \end{bmatrix}$$

$$B = \begin{bmatrix} 29 & 19 & 10 \end{bmatrix}$$

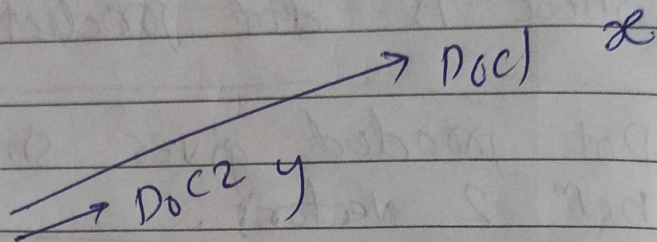
$$C = \begin{bmatrix} -30 & 100 & -10 \end{bmatrix}$$

note dot product of A & B
is high compared to AC
hence A & B are more similar
than A & C.

Doc1 $\rightarrow$ diwali at 20,000 words

Doc2 $\rightarrow$ diwali of 100 words.

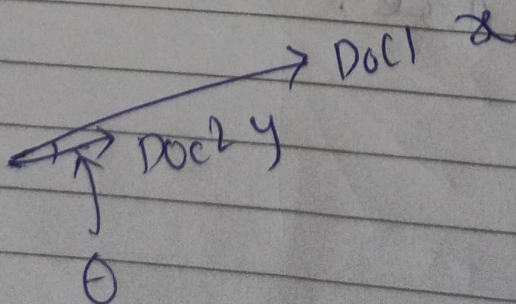
Doc1 & Doc2 can be represented in vector



both 2 docs will be similar as they describe about same thing diwali, which can be confirmed by dot product

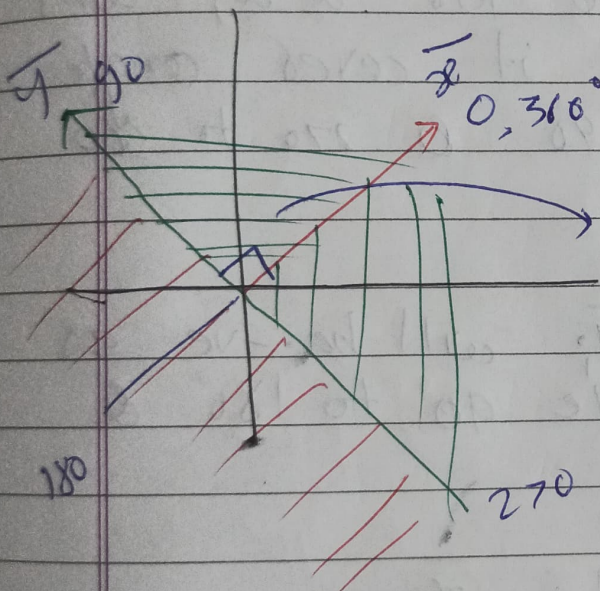
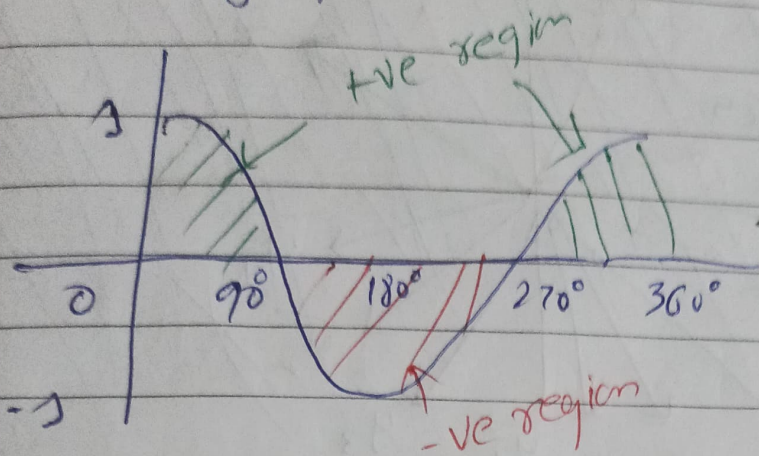
$$x^T \cdot y = \|x\| \|y\| \cos \theta$$

$$\cos \theta = \frac{x^T \cdot y}{\|x\| \|y\|}$$



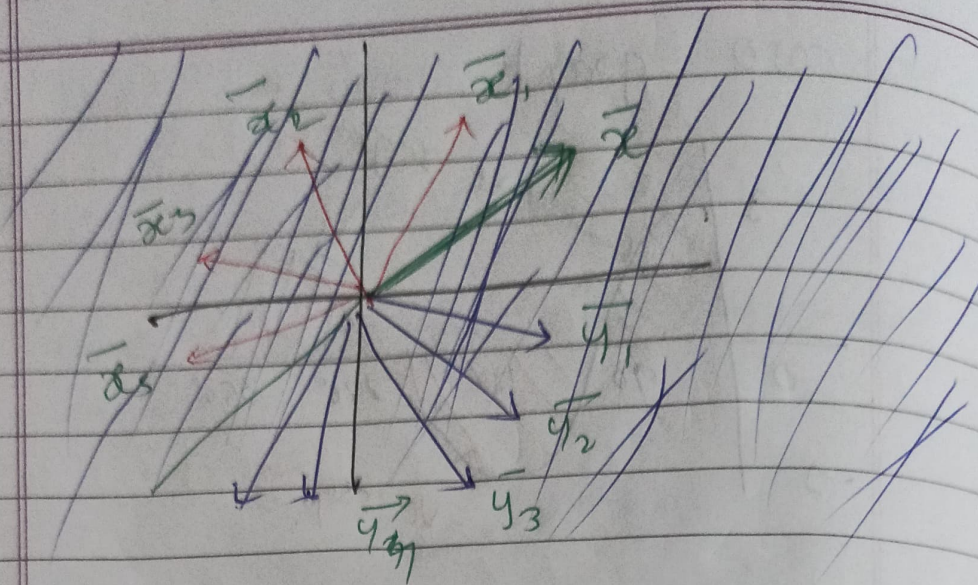
$\cos \theta$ will be very less here as x & y are similar.

$\cos \theta$ graph



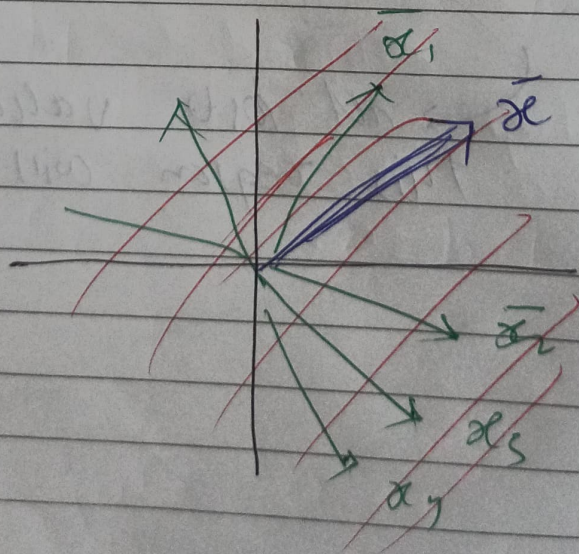
all $\cos \theta$ value
in this region
will be +ve

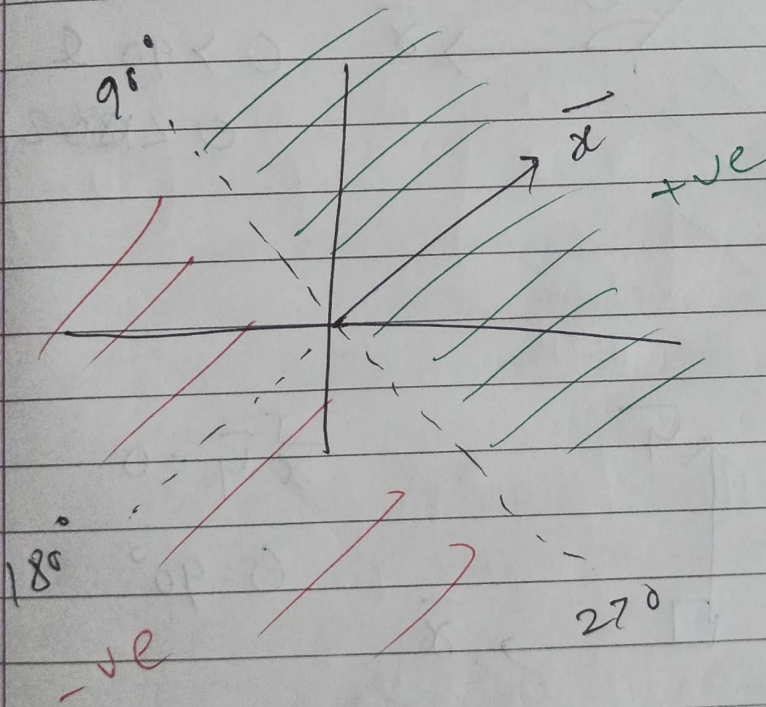
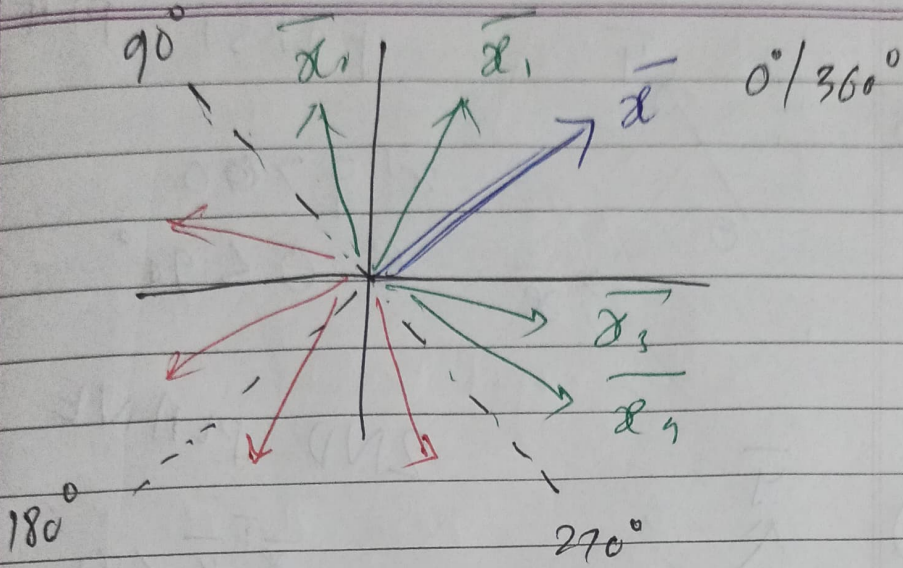
all $\cos \theta$ value in
this region will be -ve



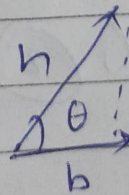
Dot prod. $\cos \theta$ for $\bar{x} \bar{x}$; will be +ve as it covers angle betⁿ 0 to 90 & 270 to 360 will be

$\cos \theta$ for $\bar{x} \bar{y}$; will be -ve as it covers angle 90° to 180° & 180° to 270



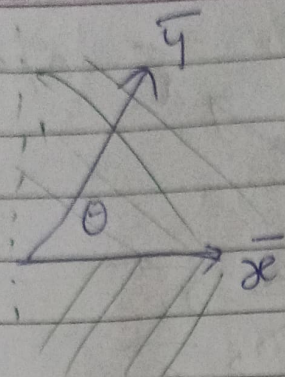


$$\cos \theta = \frac{b}{h}$$



FIRST PLANE

①

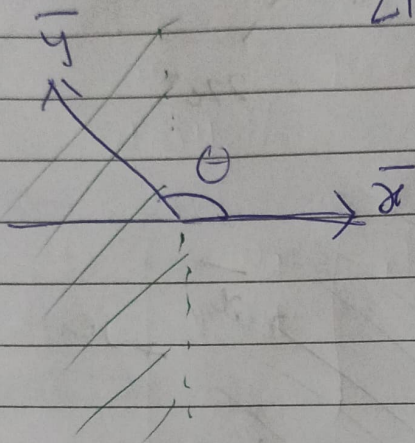


$$\bar{x}^T \bar{y} > 0$$

$$0 < \theta < 90^\circ$$

2ND PLANE

②

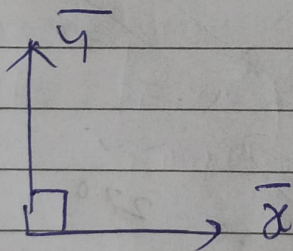


$$\bar{x}^T \bar{y} < 0$$

$$\theta > 90^\circ$$

$$\theta < 270^\circ$$

③



$$\bar{x}^T \bar{y} = 0$$

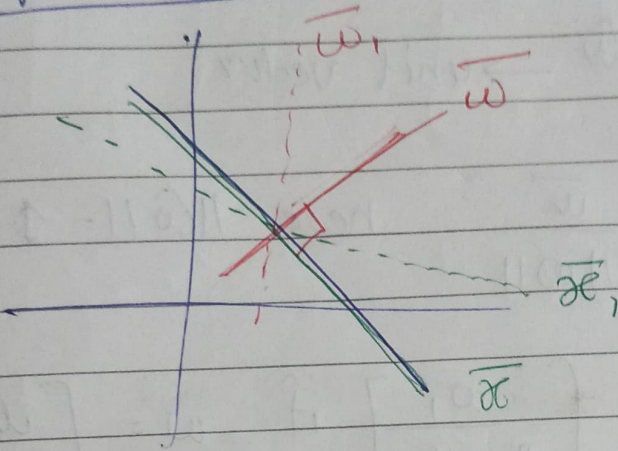
$$\theta = 90^\circ$$

① & ② are 2 half planes.

$w^T x = 0$
 ↑
 weight vector → feature vector

$$\bar{w}^T \bar{x} + w_0 = 0$$

Weight vector $\bar{w}$ is always perpendicular to feature vector $\bar{x}$



when someone changes the weight then the feature vector $\bar{x}$ changes. Show by dotted line $\bar{x}_1$ & $\bar{w}_1$.

In machine learning we need to find the line i.e. feature vector $\bar{x}$ which will divide (fit) points. We modify $\bar{x}$ with help of $\bar{w}$ so that the division of point will be optimum.

unit vector

$\vec{w}$ vector

$\hat{w}$ $\rightarrow$ unit vector

$$\hat{w} = \frac{\vec{w}}{\|\vec{w}\|} \quad \text{here } \|\hat{w}\| = 1$$

$$\hat{w} = \begin{bmatrix} \frac{w_1}{\|\vec{w}\|} \\ \frac{w_2}{\|\vec{w}\|} \end{bmatrix} \quad \text{if } \vec{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

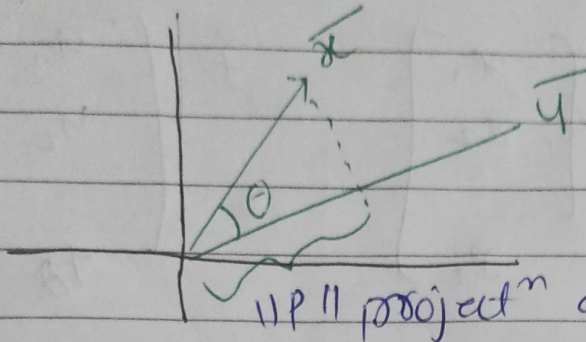
$$\|\hat{w}\| = \sqrt{\left(\frac{w_1}{\|\vec{w}\|}\right)^2 + \left(\frac{w_2}{\|\vec{w}\|}\right)^2}$$

$$= \sqrt{\frac{w_1^2 + w_2^2}{(\|\vec{w}\|)^2}}$$

$$= \sqrt{\frac{w_1^2 + w_2^2}{\left(\sqrt{w_1^2 + w_2^2}\right)^2}}$$

$$= \sqrt{\frac{w_1^2 + w_2^2}{w_1^2 + w_2^2}} = 1$$

Projectⁿ of Vector



$\|p\|$ projectⁿ of $\vec{x}$ on $\vec{y}$

$$\cos \theta = \frac{\vec{x}^T \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|}$$

$$\cos \theta = \frac{\text{base}}{\text{hypotenuse}} = \frac{\|p\|}{\|\vec{x}\|}$$

$$\frac{\|p\|}{\|\vec{x}\|} = \frac{\vec{x}^T \vec{y}}{\|\vec{x}\| \|\vec{y}\|}$$

$$\|p\| = \frac{\vec{x}^T \vec{y}}{\|\vec{y}\|}$$

$$\|p\| = \vec{x}^T \hat{y}$$

Recap
vectors $\bar{x}, \bar{y} \in \mathbb{R}^d$

$$\bar{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \quad \bar{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_d \end{bmatrix}$$

② norm $\|\bar{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_d^2}$

③ dot product $= \bar{x}^T \bar{y} = \sum_{i=1}^d x_i y_i$

④ Angle betⁿ 2 vectors

$$\cos \theta = \frac{\bar{x}^T \bar{y}}{\|\bar{x}\| \|\bar{y}\|}$$

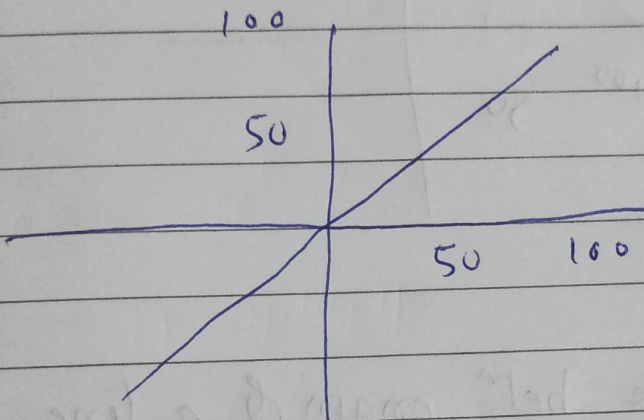
Shifting of line

$$w_1x + w_2y + w_0 = 0$$

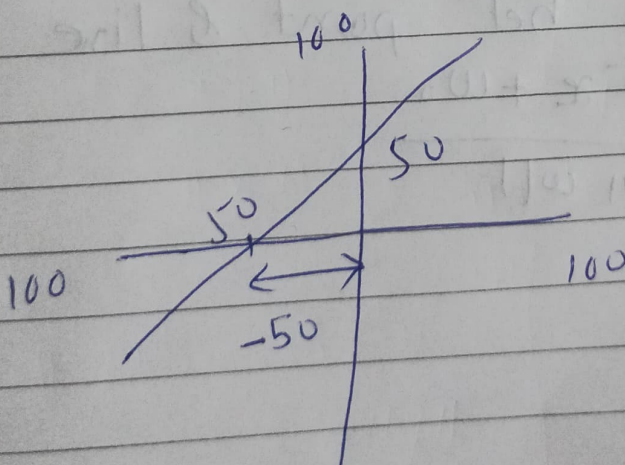
$$w_1 = -4.3$$

$$w_2 = 4.3$$

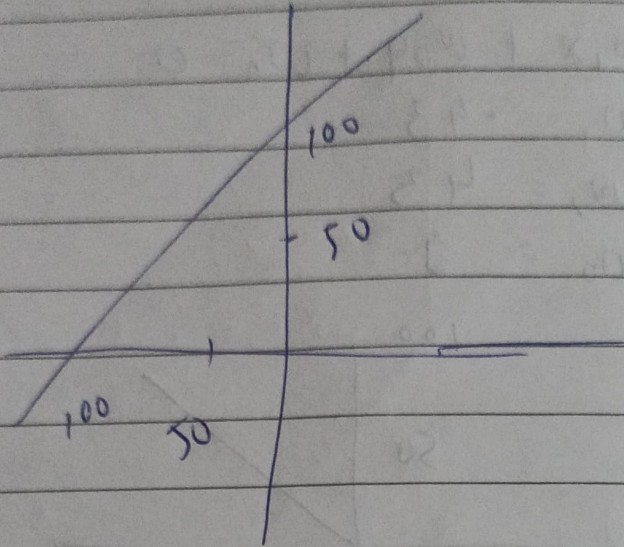
$$w_0 = 1$$



$$w_1(x+50) + w_2y + w_0 = 0$$



$$w_1(x+50) + w_2(y-50) + w_0 = 0$$



Distance betⁿ origin & a line

$$\frac{-w_0}{\|w\|}$$

Distance betⁿ point & line

$$\frac{w^T x + w_0}{\|w\|}$$